

Exercises 1.5

1. (a) The automorphisms of C_{16} are given by $g \mapsto g, g \mapsto g^3, g \mapsto g^5, g \mapsto g^7, g \mapsto g^9, g \mapsto g^{11}, g \mapsto g^{13}, g \mapsto g^{15}$.
 (b) There are $\varphi(84) = 24$ automorphisms of C_{84} .
2. $(\mathbb{Z}, +)$ is generated by 1 and by -1 . An automorphism ψ of $\mathbb{Z}$ is determined by its effect on 1, since $\psi(n) = \psi(n \cdot 1) = n \cdot \psi(1)$ for all $n \in \mathbb{Z}$.
 ψ must map the generator 1 to a generator, so $\psi(1) = 1$ or $\psi(1) = -1$. The former is the identity map. For the latter, we have $\psi(m+n) = (m+n) \cdot (-1) = m \cdot (-1) + n \cdot (-1) = \psi(m) + \psi(n)$, so ψ is a group homomorphism and is clearly bijective. Thus there are just two automorphisms. $\text{Aut}(\mathbb{Z}) \cong C_2$.
3. Let $\phi : A \mapsto A^{-1}$. Then $\phi(AB) = (AB)^{-1} = B^{-1}A^{-1}$, but $\phi(A)\phi(B) = A^{-1}B^{-1}$. These are not equal in general, so ϕ is not an automorphism.
 (It's certainly a bijective map. In fact it is an *anti-automorphism*.)
4. (a) For any $x \in G$, $\gamma_b^{-1}(x) = y$ where $\gamma_b(y) = x$, i.e. $byb^{-1} = x$, so $y = b^{-1}xb$.
 Thus $\gamma_a\gamma_b^{-1}(x) = \gamma_a(b^{-1}xb) = ab^{-1}xba^{-1} = (ab^{-1})x(ab^{-1})^{-1} = \gamma_{ab^{-1}}(x)$, so $\gamma_a\gamma_b^{-1} = \gamma_{ab^{-1}}$.
 Hence $\gamma_a\gamma_b^{-1} \in \text{Inn}(G)$ for all $\gamma_a, \gamma_b \in \text{Inn}(G)$ so, by the subgroup test, $\text{Inn}(G)$ is a subgroup of $\text{Aut}(G)$.
 Let ϕ be any automorphism of G . For each $x \in G$, $\phi\gamma_a\phi^{-1}(x) = \phi(a\phi^{-1}(x)a^{-1}) = \phi(a)\phi(\phi^{-1}(x))\phi(a^{-1}) = \phi(a)x\phi(a)^{-1} = \gamma_{\phi(a)}(x)$.
 Thus $\phi\gamma_a\phi^{-1} = \gamma_{\phi(a)}$, so $\phi\gamma_a\phi^{-1} \in \text{Inn}(G)$ for all $\phi \in \text{Aut}(G)$ and all $\gamma_a \in \text{Inn}(G)$.
 Thus $\text{Inn}(G) \triangleleft \text{Aut}(G)$.
 (b) Taking b^{-1} for b in the first result from (a) shows that $\gamma_{ab} = \gamma_a\gamma_b$, so $\psi(ab) = \psi(a)\psi(b)$ and hence ψ is a group homomorphism.
 $\text{Ker } \psi$ is the set of all $a \in G$ for which γ_a is the identity map, i.e. $axa^{-1} = x$ for all $x \in G$. Thus $\text{Ker } \psi = \{a \in G : ax = xa \text{ for all } x \in G\} = Z(G)$.
 The image of ψ is $\text{Inn}(G)$. By the first isomorphism theorem, $G/Z(G) \cong \text{Inn}(G)$.
 If $Z(G)$ is trivial then $G/Z(G) \cong G$, so $\text{Inn}(G)$ is a subgroup of $\text{Aut}(G)$ that is isomorphic to G .
5. (a) $\phi(\sigma)$ has order $o(\sigma) = 3$, so $\phi(\sigma)$ can only be σ or σ^2 .
 $\phi(\tau)$ has order $o(\tau) = 2$, so $\phi(\tau)$ can only be $\tau, \sigma\tau$ or $\sigma^2\tau$.
 Thus the maximum number of possible automorphisms is $2 \times 3 = 6$.
 (b) By inspection of the table for S_3 we see that only ι commutes with all elements, so $Z(S_3)$ is trivial. Hence by 4(b) $\text{Aut}(S_3)$ has a subgroup $\text{Inn}(S_3)$ isomorphic to S_3 . But by 5(a) $|\text{Aut}(S_3)| \leq 6 = |S_3|$, so $\text{Aut}(S_3) \cong S_3$.
6. Let $H = \langle \sigma \rangle \cong C_n$ and $J = \langle \tau \rangle \cong C_2$. These are subgroups of D_n , and $H \triangleleft D_n$ since $[D_n : H] = 2$.
 $|H| = n$ and $|J| = 2$. As $\tau \notin H$ we have $H \cap J = \{\iota\}$, so $|HJ| = 2n$, so $D_n = HJ$. Thus $D_n = H \rtimes J$.
 (Note that if n is even then, generalising Exercises 1.3 Question 8, we can also express D_n as a direct product of subgroups isomorphic to $D_{n/2}$ and C_2 .)

7. Let $A = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$, $B = \begin{pmatrix} \omega & 0 \\ 0 & \omega^2 \end{pmatrix}$. Then $A^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$,
 $A^3 \neq I_2$, $A^4 = I_2$ so A has order 4, $B^2 = \begin{pmatrix} \omega^2 & 0 \\ 0 & \omega \end{pmatrix}$, $B^3 = I_2$ so B has order 3,
and $BAB = \begin{pmatrix} 0 & \omega i \\ \omega^2 i & 0 \end{pmatrix} \begin{pmatrix} \omega & 0 \\ 0 & \omega^2 \end{pmatrix} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} = A$.
8. As H and J have orders coprime orders, $|H \cap J| = 1$ so $|HJ| = 20$, hence $G = HJ = \{h^r j^s : h \in \{0, 1, 2, 3, 4\}, j \in \{0, 1, 2, 3\}\}$. As $H \triangleleft G$, we have $G = H \rtimes J$.
 $j h j^{-1}$ must be a generator of H so $j h j^{-1} = h^r$ where r is coprime to 5.
For the given matrices X and Y over $\mathbb{F}_5$, we find that $X^5 = Y^4 = I_2$ and $YXY^{-1} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = X^2$, i.e. $r = 2$.
9. Let h, j be generators of H and J . If $H \triangleleft G$ then $j h j^{-1} = h^r$ for some r .
Then $h = j^3 h j^{-3} = h^{(r^3)}$, so $r^3 \equiv 1 \pmod{8}$. By evaluating $1^3, \dots, 7^3$ modulo 8 we find that $r = 1$, so $j h = h j$, so G is abelian (isomorphic to C_{24}).
If $J \triangleleft G$ then the same reasoning gives $h j h^{-1} = j^r$ where $r^8 \equiv 1 \pmod{3}$, so $r \in \{1, 2\}$.
If $r = 1$ then G is abelian as above.
If $r = 2$ then $G = \langle h, j : h^8 = j^3 = e, j h j = h \rangle$. Letting $\omega = e^{2\pi i/8}$, $\eta = e^{2\pi i/8}$,
 $X = \begin{pmatrix} \omega & 0 \\ 0 & \omega^2 \end{pmatrix}$, $Y = \begin{pmatrix} 0 & \eta \\ \eta & 0 \end{pmatrix}$, we find that $X^3 = Y^8 = I_2$ and $XYX = Y$, so G is isomorphic to the subgroup of $M_2(\mathbb{C})$ generated by $\{X, Y\}$.
10. (a) Let $G = \langle g \rangle$. Every subgroup of G is cyclic. Let $H = \langle g^r \rangle$. We know that any automorphism of G has the form $\phi : g \mapsto g^m$, so $\phi(g^r) = g^{mr} = (g^r)^m$. Thus $\phi(H) \subset H$. As ϕ is bijective, $\phi(H) = H$ so H is characteristic in G .
(b) If H is characteristic then $\gamma_a(H) = H$ for every inner automorphism γ_a of G , so H is closed under conjugation by all elements of G , i.e. $H \triangleleft G$.
(c) Let ϕ be an automorphism of G , so $\phi(H) = H$. Thus ϕ (restricted to J) is an automorphism of H , so $\phi(J) = J$ and hence J is characteristic in G .
If we only know that $J \triangleleft H \triangleleft G$ then $\gamma_a(H) = H$ for every inner automorphism γ_a of G , but γ_a (restricted to H) is not in general an inner automorphism of H (i.e. it is not γ_h for some $h \in H$) so we cannot deduce that J is closed under conjugation by all elements of G .
11. By Proposition 1.8, G has a subgroup Q isomorphic to G/N . Now $|N|, |Q|$ are coprime so $|N \cap Q| = 1$ and $|N| \cdot |Q| = |N| \cdot |G|/|N| = |G|$, so $NQ = G$. Thus $G = N \rtimes Q$.
12. $(h_1, j_1)((h_2, j_2)(h_3, j_3)) = (h_1, j_1)(h_2 \theta_2(h_3), j_2 j_3) = (h_1 \theta_1(h_2 \theta_2(h_3)), j_1 j_2 j_3)$.
 $((h_1, j_1)(h_2, j_2))(h_3, j_3) = (h_1 \theta_1(h_2), j_1 j_2)(h_3, j_3) = (h_1 \theta_1(h_2) \theta(j_1 j_2)(h_3), j_1 j_2 j_3)$.
Now $\theta(j_1 j_2) = \theta(j_1) \theta(j_2) = \theta_1 \theta_2$, so $h_1 \theta_1(h_2) \theta(j_1 j_2)(h_3) = h_1 \theta_1(h_2) \theta_1 \theta_2(h_3) = h_1 \theta_1(h_2 \theta_2(h_3))$.
Thus $(h_1, j_1)((h_2, j_2)(h_3, j_3)) = ((h_1, j_1)(h_2, j_2))(h_3, j_3)$, i.e. the operation is associative.
 $(h_1, j_1)(e_H, e_J) = (h_1 \theta_1(e_H), j_1 e_J) = (h_1 e_H, j_1 e_J) = (h_1, j_1)$
and $(h_1, j_1)(\theta_1^{-1}(h_1^{-1}), j_1^{-1}) = (h_1 \theta_1(\theta_1^{-1}(h_1^{-1})), j_1 j_1^{-1}) = (h_1 h_1^{-1}, j_1 j_1^{-1}) = (e_H, e_J)$, so we have a left identity and left inverses which (as shown in Groups & Rings) must also be a right identity and right inverses. Thus $H \rtimes_{\theta} J$ is a group.
When $\theta(j)$ is the identity map for all $j \in J$, we get the direct product $H \times J$.

For $\overline{H}$ and $\overline{J}$ as defined, clearly $\overline{H} \cap \overline{J} = \{(e_H, e_J)\}$, the identity of $H \rtimes_{\theta} J$, and $\overline{H}\overline{J} = \{(h, j) : h \in H, j \in J\} = H \times J$.

Furthermore, $(e_H, j)(h, e_J)(e_H, j)^{-1} = (\theta(j)(h), j)(e_H, j^{-1}) = (\theta(j)(h), e_J) \in \overline{H}$ since $\theta(j) : H \rightarrow H$, so $\overline{H} \triangleleft H \rtimes_{\theta} J$. Hence $H \rtimes_{\theta} J = \overline{H} \rtimes \overline{J}$.

13. D_4 has centre $\{\iota, \sigma^2\}$ so $|G/\text{Inn}(G)| = |Z(G)| = 2$, hence $|\text{Inn}(G)| = 4$, i.e. there are 4 inner automorphisms. They are determined by their effect on the generators σ and τ .

D_4 has two elements of order 4 and five elements of order 2. σ and τ must map to elements of the same order, so the maximum possible number of automorphisms is $2 \times 5 = 10$. Also the order of $\text{Aut}(G)$ is divisible by that of its subgroup $\text{Inn}(G)$, so it is either 4 or 8.

Consider the map ϕ defined on the generators by $\sigma \mapsto \sigma$ and $\tau \mapsto \sigma\tau$. We can check that this is an automorphism but is not one of the inner automorphisms.

Thus $\text{Aut}(D_4)$ has order 8. ϕ does not commute with all the inner automorphisms, so $\text{Aut}(D_4)$ is non-abelian. Conjugation by σ^2 and by τ are two distinct inner automorphisms of order 2, so $\text{Aut}(D_4)$ is not isomorphic to Q_4 . Hence $\text{Aut}(D_4) \cong D_4$.

(See this StackExchange question for more detail.)